FORECAST PROJECT ON HEATING OIL IN U.S. MARKET

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Time Series Analysis module - prof. Ravazzolo Francesco

00 - HEATING OIL

WHAT IS IT?

A refined product **from crude oil** mainly used in **space heating**

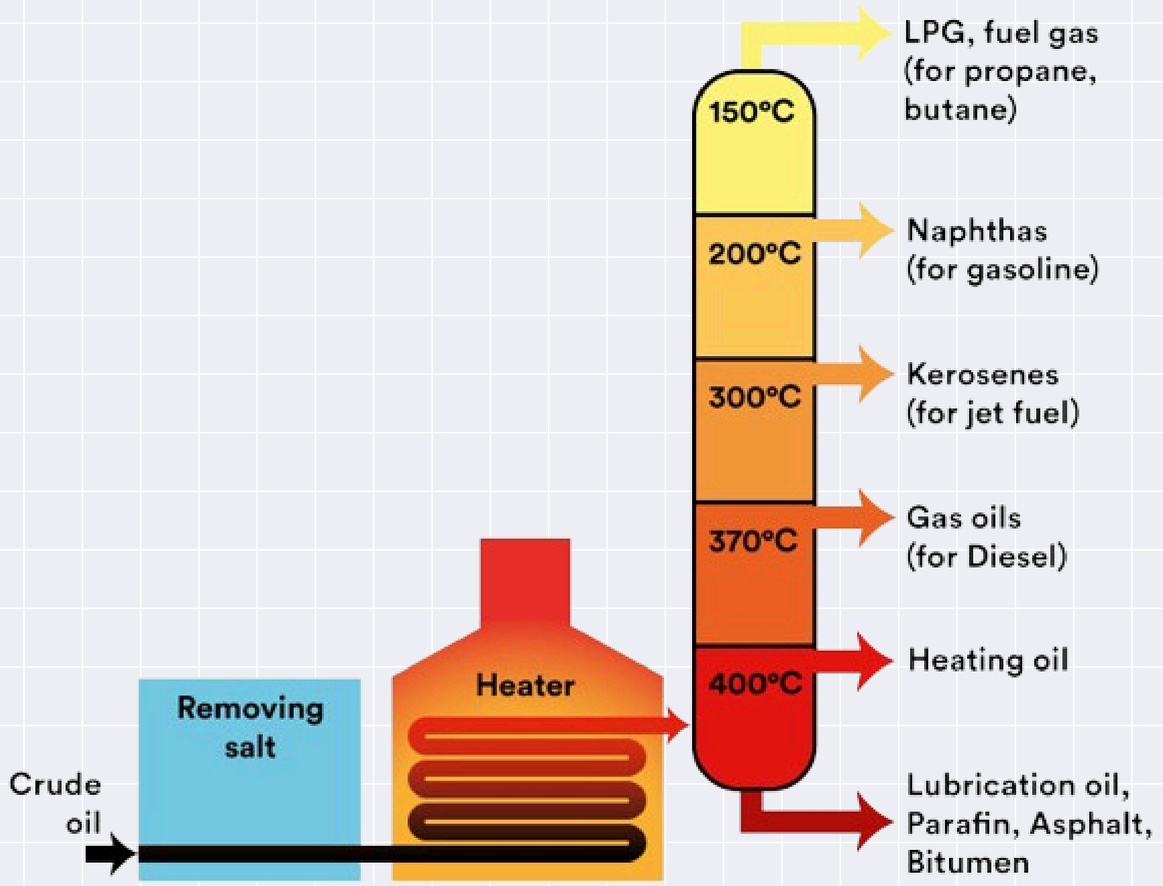
WHERE IS IT USED?

Used in:

- **residencial heating,** especially in Northeast States of USA
- **Commercial buildings** (schools, offices...)

WHY DO ITS PRICES MATTER?

- Household energy costs
- Inflation measure
- **Energy markets**
- Environmental policy



00 - WORKFLOW SUMMARY

01 **Preliminary analysis** → Data display sources of **non-stationarity**

02 **ARIMA modeling** → Using ACF and PACF, possible **ARIMA model** was identified

03 **GARCH-type modeling** → **GARCH-type models** were identified to handle **volatility**

04 **Models comparison in forecasting** → **Forecasting performance** was compared using **evaluation metrics**

01 - PRELIMINARY ANALYSIS

ORIGINAL TIME SERIES' TRANSFORMATIONS

- **40 year-period of daily observations**
- **variability** in levels

↓

monthly aggregation

+

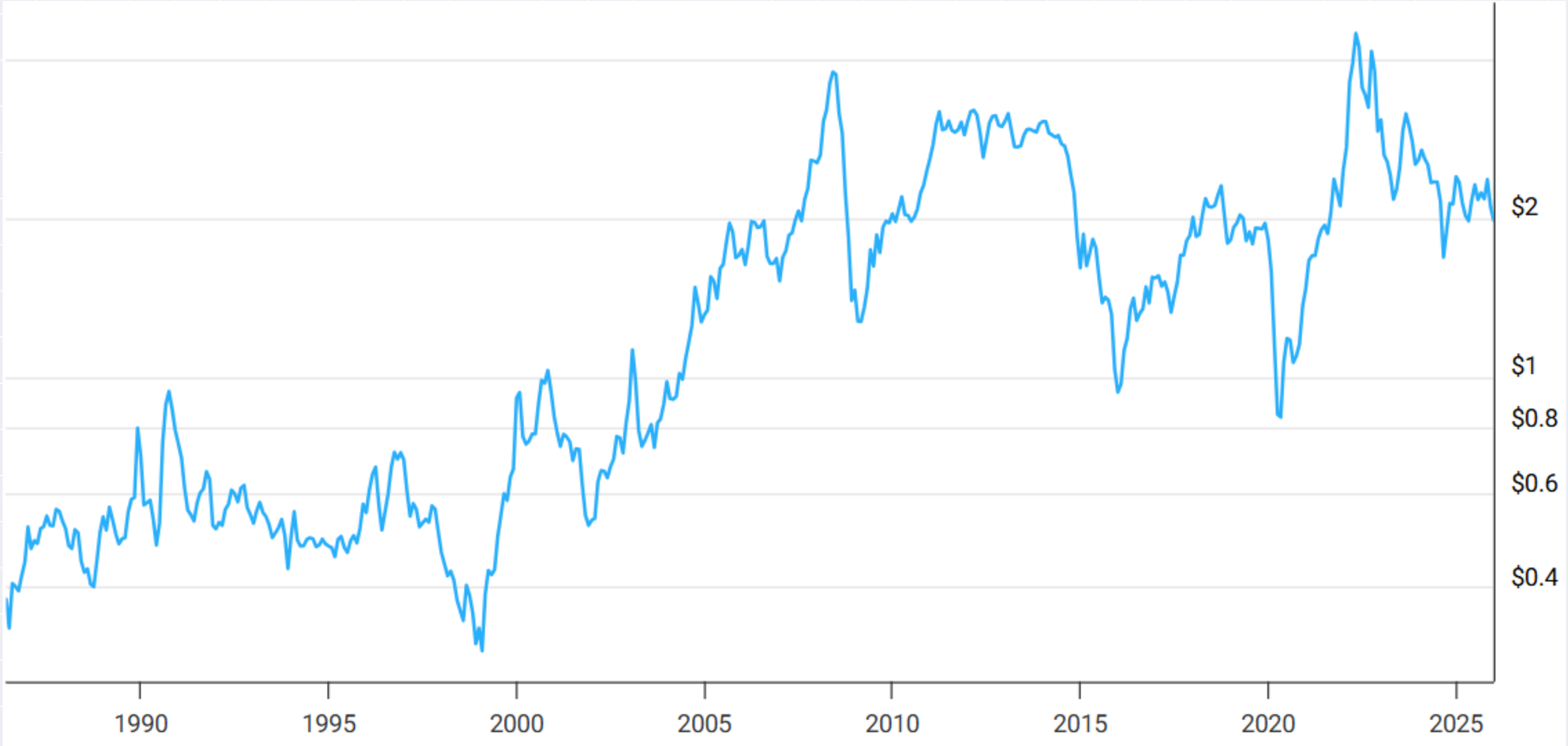
logarithmic transformation

↓

improved stationarity

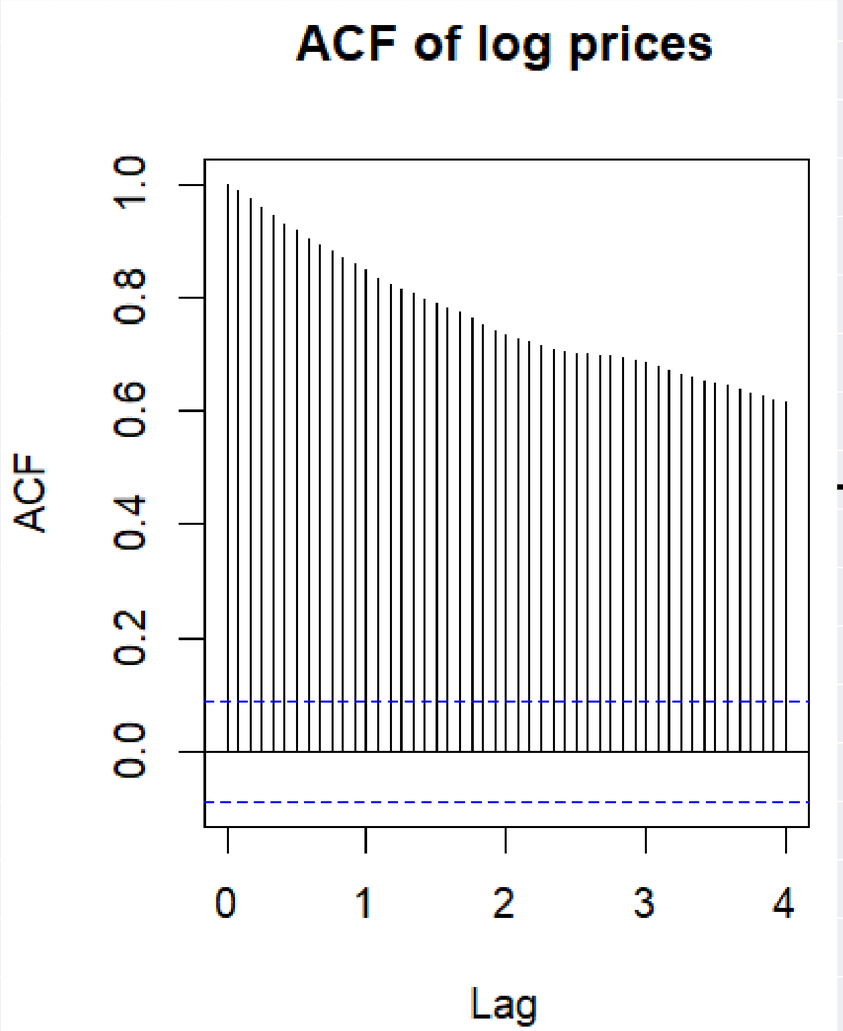
+

log - returns

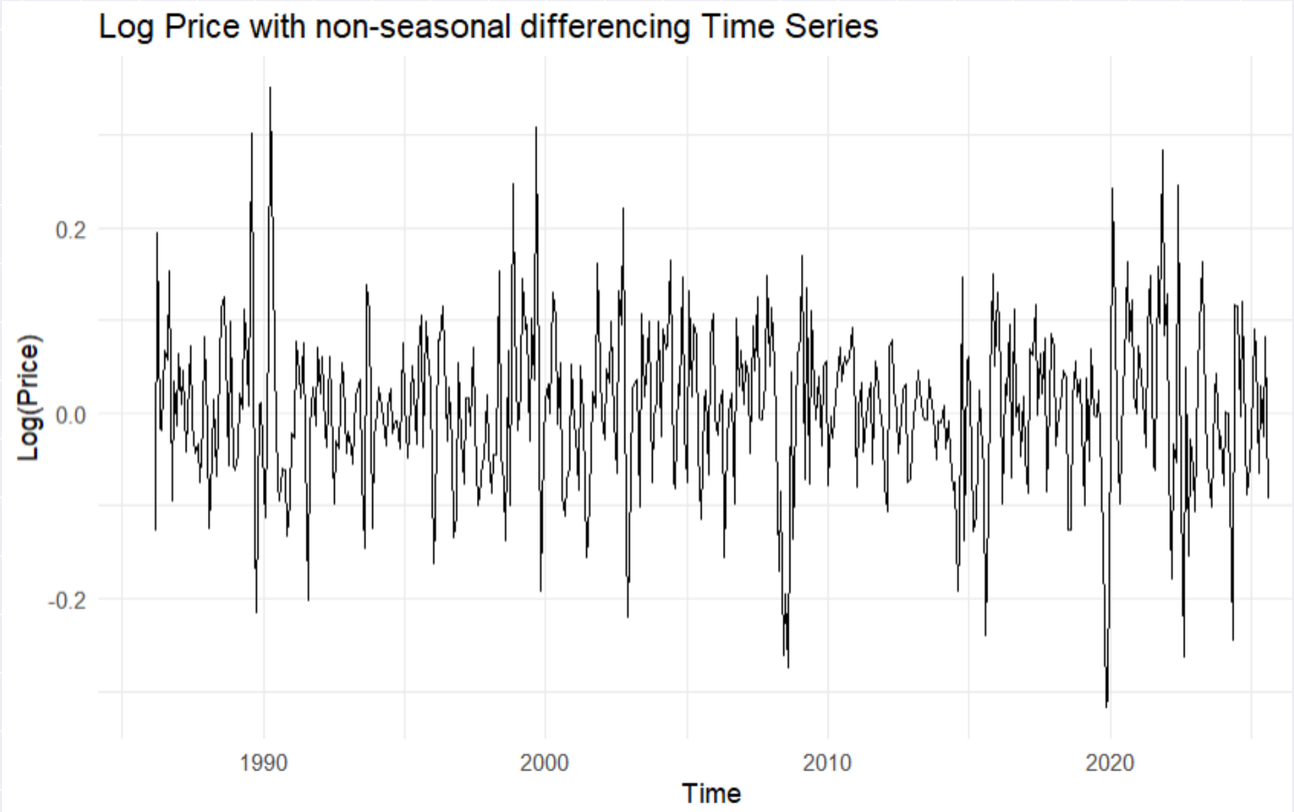


01 - PRELIMINARY ANALYSIS

SOURCES OF NON -STATIONARITY IDENTIFICATION



∇ of order 1 →



- strong linear temporal dependence

- randomly flactuates around a **costant mean**
- **no** observable **trend**

01 - PRELIMINARY ANALYSIS

SEASONALITY CONCERNS

Reasons why the seasonal component **can't be detected**

1 Heating oil **follows crude** and refining economics

2 **Storage spreads** seasonal effects over the months

3 **national level**

- 3a **Mixed demand** (heating + transaportation)
- 3b **different climates aggregation**

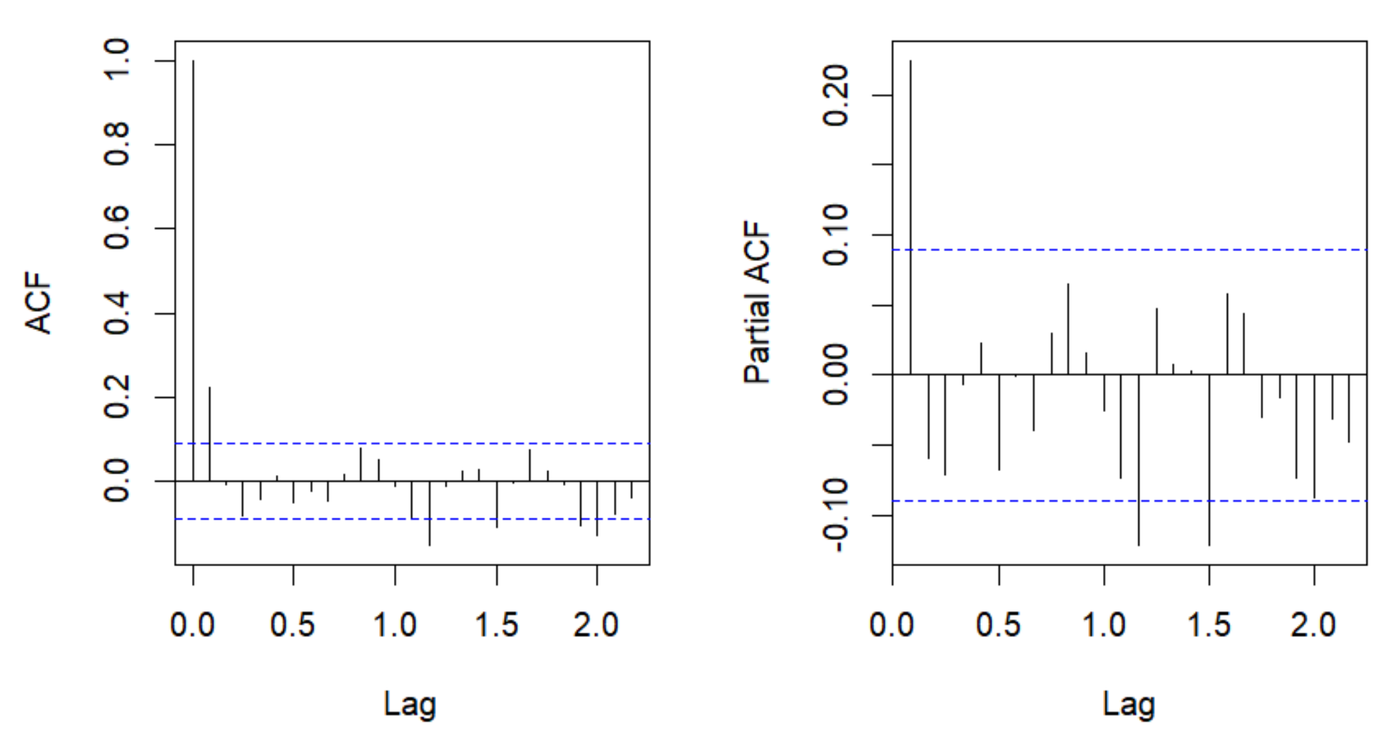
02 - MODEL IDENTIFICATION

- ARIMA(1,1,1)
- ARIMA(1,1,0) - more parsimonious

→ to choose:

- **coeftest()**
- **AIC**
- **BIC**

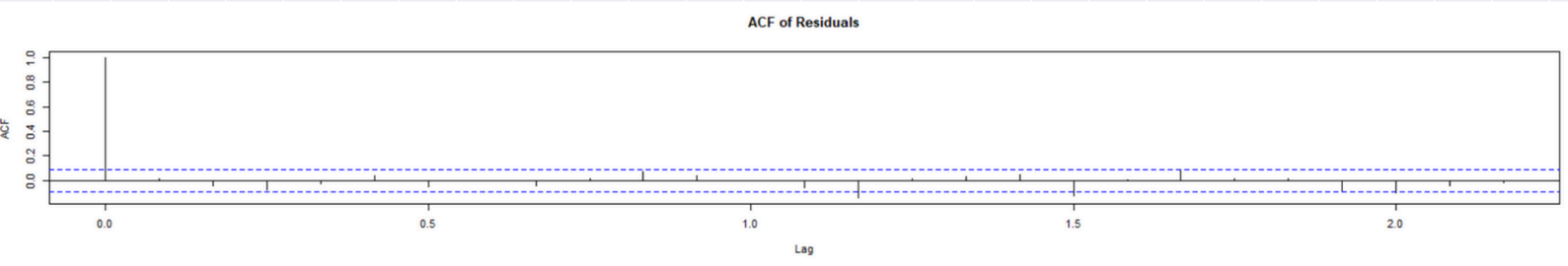
→ **all** suggest ARIMA(1,1,0)



	TERM	ESTIMATES	P-VALUES	AIC	BIC
ARIMA(1,1,0)	AR1	0.22711546	4.030279e-07	-964.2954	-955.9730
ARIMA(1,1,1)	AR1	0.07245596	6.509960e-01	-963.4124	-950.9288
ARIMA(1,1,1)	MA1	0.16649603	2.875490e-01		

02 - MODEL IDENTIFICATION

RESIDUAL ANALYSIS ON CHOSEN ARIMA(1,1,0)



→ no autocorrelation structure

	LJUNG-BOX	JARQUE-BERA	ARCH LM
ARIMA(1,1,0)	0.5544945	4.520847e-07	0.0003011959

- Jarque-Bera: residuals **deviate from normality**
- ARCH LM: **heteroskedasticity** → GARCH-type required

03 - GARCH-TYPE MODELING

- Volatility as a dynamic process
- Capture volatility clustering

DATASET PREPARATION

- Monthly aggregated data
- Differentiated log-returns
- **Winsorization** at 5th and 95th percentile

BENCHMARK FOR GARCH MODELS

- Fitted a **sGARCH(1,1)** + ARMA(1,0)
- **No** remaining **ARCH effects** (0.1959)
- **asymmetryc response to shocks is captured**

p < 0.05 → statistically significant
p ≥ 0.05 → NOT significant

	prob	sign
Sign Bias	0.07163432	*
Negative Sign Bias	0.63730643	
Positive Sign Bias	0.63320528	
Joint effect	0.16431248	

03 - GARCH-TYPE MODELING

EGARCH

- models log-volatility and **naturally incorporates asymmetry**
- negative shocks have a stronger impact than positive ones

APARCH

- Introduces a flexible **power parameter** and **leverage term**
- **Variance can react differently to magnitude and sign of shocks**

Fitting EGARCH / APARCH (1,1) + ARMA(1,0)

- **Sign-bias diagnostics not statistically significant**
- **No** remaining **ARCH effects** (0.2187 / 0.2267)

↓

most of the asymmetry and heteroskedasticity is captured

	prob EGARCH	prob APARCH	sign
Sign Bias	0.0572833	0.06775571	*
Negative Sign Bias	0.3781710	0.06775571	
Positive Sign Bias	0.4945227	0.06775571	
Joint effect	0.2343415	0.06775571	

03 - GARCH-TYPE MODELING

VALIDATE ALL GARCH MODELS

	LjungBox	ARCH LM TEST	Jarque-Bera
SGARCH(1,1)-ARMA(1,0)	0.6936489	0.1958935	0.04377816
EGARCH(1,1)-ARMA(1,0)	0.6891787	0.2187179	0.03915143
APARCH(1,1)-ARMA(1,0)	0.6919096	0.2266875	0.04240270

- All seem **statistically adequate**
- As financial returns: **deviation from normality** but a skewed Student-t is already assumed

04 - FORECASTING: rolling window simulations

WHAT WE DID

- Conducted a rolling-window out-of-sample analysis on monthly data on each candidate model
- Used a 10-year fixed estimation window, rolled forward one month at a time
- Re-estimated the models at each step using only past information
- Generated one-step-ahead forecasts and compared them with realized returns

CANDIDATE MODELS:

SGSRCH(1,1)+ARMA(1,0)
APARCH(1,1)+ARMA(1,0)
EGARCH(1,1)+ARMA(1,0)
vs
RANDOM WALK

WHY WE DID

- To test true forecasting ability, not just in-sample fit
- To mimic real-time forecasting, with time-varying parameters
- Allows evaluation of accuracy, stability, and robustness over time
- To test our candidate models and test them against a random walk

04 - FORECASTING: point forecast accuracy

RMSE	Measures the average magnitude of forecast errors, giving more weight to large errors. Lower values indicate better forecast accuracy.
MAE	Measures the average absolute forecast error. It is less sensitive to outliers than RMSE and easier to interpret.

	SGARCH_11	APARCH_10	EGARCH_10	RW
RMSE	0.4310666	0.4311562	0.4317265	0.6056833
MAE	0.3541980	0.3541099	0.3548882	0.5053200

RESULTS

→ GARCH models substantially outperform the random walk benchmark
→ Differences among GARCH-type models are small, with a slight edge for APARCH(1,0)
→ Modeling conditional heteroskedasticity adds clear value for forecasting return

04 - FORECASTING: point forecast accuracy

DM p-value	Tests whether the difference in forecast accuracy between two models is statistically significant. A low p-value implies that one model outperforms the other in a statistically meaningful way.

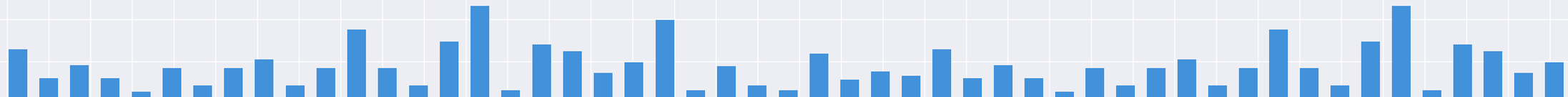
	SGARCH_11	APARCH_10	EGARCH_10	RW
SGARCH_11	NA	8.983534e-01	4.048137e-01	4.767579e-06
APARCH_10	8.983534e-01	NA	1.551525e-01	5.185769e-06
EGARCH_10	4.048137e-01	1.551525e-01	NA	5.517456e-06
RW	4.767579e-06	5.185769e-06	5.517456e-06	NA

H0: 2 models have same forecast accuracy

p-value <0.05 reject H0
p-value >0.05 do not reject

RESULTS

→ All GARCH-type models significantly outperform the Random Walk
→ All P-values are >0.05, so no model appears significantly better at this stage.



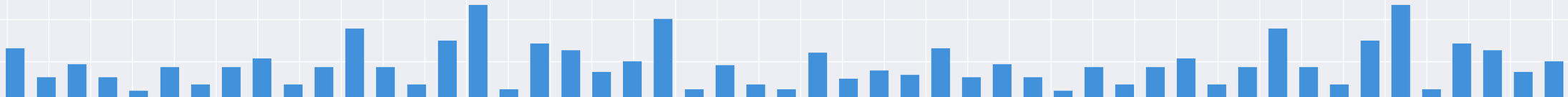
04 - FORECASTING: distributional accuracy

Log-score	Measures how much probability the predictive density assigns to the realized outcome. Lower values indicate that the model places higher likelihood on what actually occurs.
CRPS	Measures the distance between the predictive distribution and the realized outcome, capturing both calibration and sharpness, with lower values indicating better probabilistic accuracy.

Model <chr>	Mean_LogScore <dbl>	Mean_CRPS <dbl>
SGARCH_11	0.6077711	0.2492542
APARCH_10	0.6044415	0.2488251
EGARCH_10	0.6024727	0.2491714

RESULTS

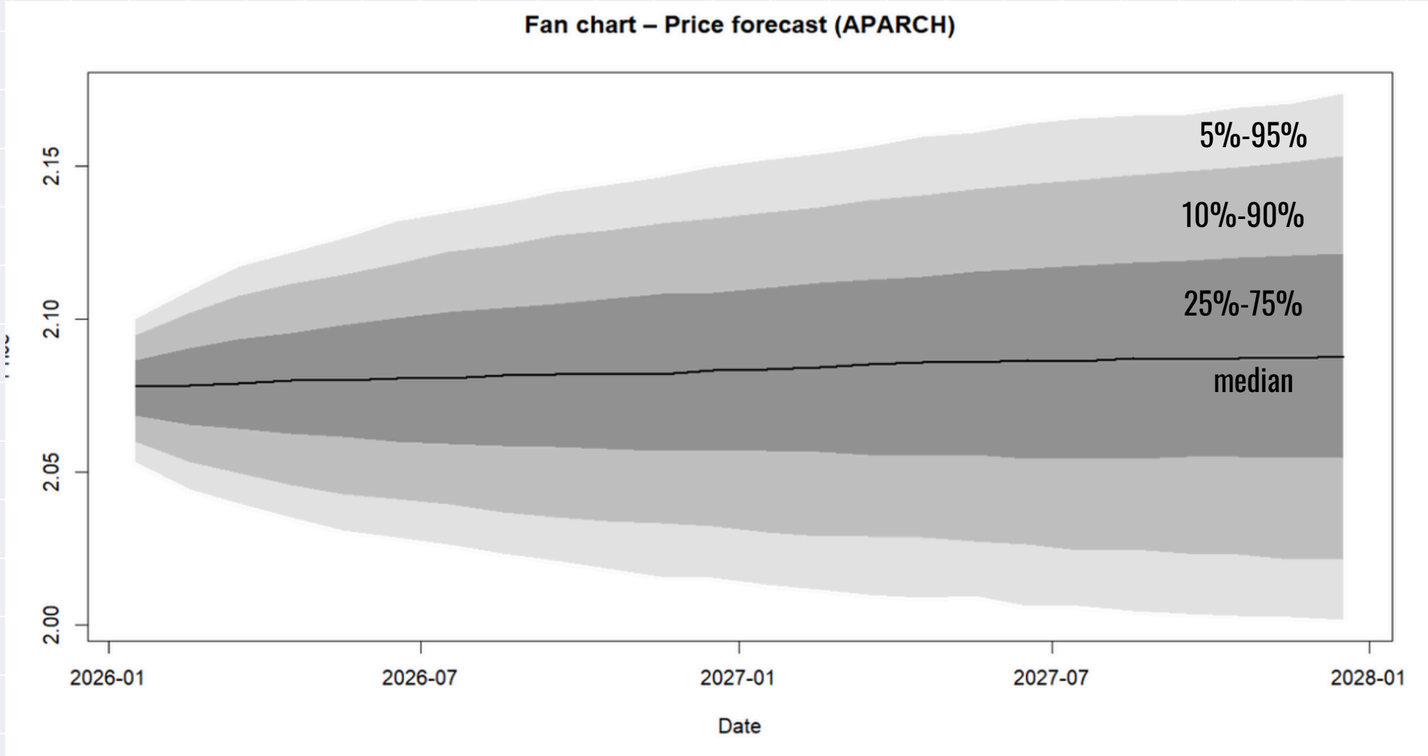
→ Lower values indicate better probabilistic forecasts
→ Differences across models are small
→ SGarch performs marginally worse



04 - FORECASTING: fan-chart

WHAT WE DID

- Fan chart on APARCH(1,1)-ARMA(1,0)
- Simulated 5000 future trajectories over 24-month
- converted returns into price levels



WHY WE DID IT

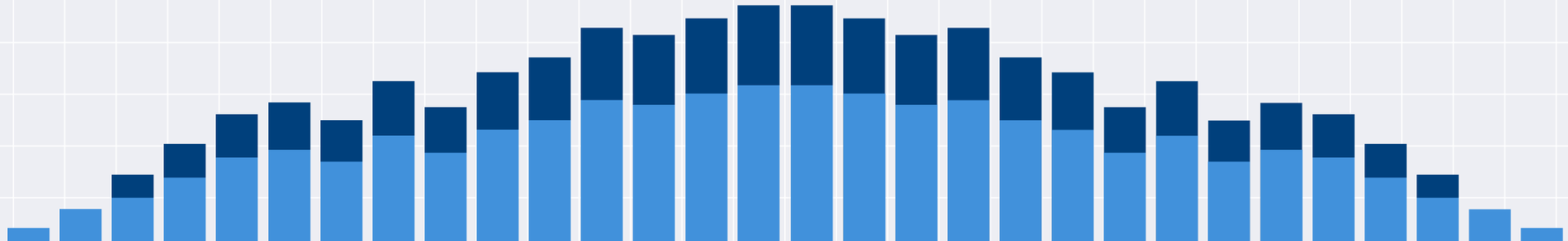
- dynamic and probabilistic view of future price evolution

RESULTS

- median forecast is stable, suggesting no strong expected price trend
- uncertainty increases over time
- model produces economically plausible density forecast

CONCLUSIONS

- GARCH-type models outperform the random walk in out-of-sample RMSE and MAE
- Diebold–Mariano tests confirm statistically superior forecasts versus the benchmark
- No significant differences among SGARCH, APARCH, and EGARCH — all perform similarly
- Log score and CRPS support the same ranking in probabilistic accuracy
- Best practical choice: simpler asymmetric models (APARCH(1,0) or EGARCH(1,0))
- Practical implication: models are credible for short-term forecasting and risk analysis, provided they are re-estimated regularly



REFERENCES

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- Seasonality concerns:

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  https://www.princeton.edu/~deaton/downloads/Competitive_Storage_and_Commodity_Price_Dynamics.pdf

  https://www.eia.gov/finance/markets/products/balance.php

THANKS FOR YOUR ATTENTION!

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Time Series Analysis module - prof. Ravazzolo Francesco